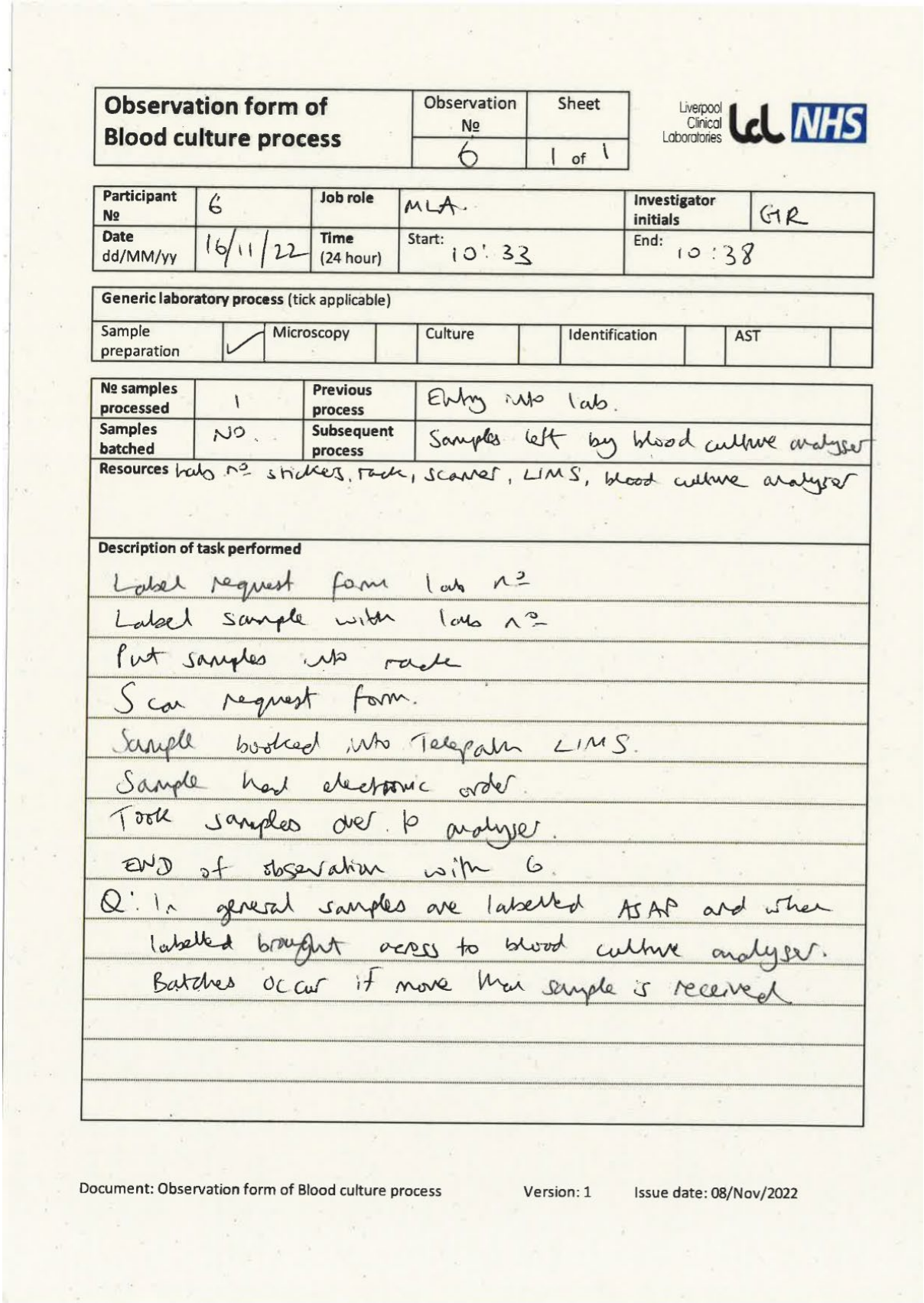
Supplementary material: Simulation to optimize the laboratory diagnosis of bacteremia

Table of Contents

Observation form example (Figure S1) *2*
Simulation input parameters (Table S1) *3*
Blood culture process maps (Figure S2–Figure S5) *4*
Supplementary plots (Figure S6–Figure S8) *8*

Observation form example (Figure S1)



Observation form of Blood culture process

Observation № 6 | Sheet 1 of 1

Liverpool Clinical Laboratories — NHS

Participant №	6	Job role	MLA.	Investigator initials	GR
Date dd/MM/yy	16/11/22	Time (24 hour)	Start: 10:33	End: 10:38	

Generic laboratory process (tick applicable)

Sample preparation	✓	Microscopy		Culture		Identification		AST	

№ samples processed	1	Previous process	Entry into lab.
Samples batched	NO.	Subsequent process	Samples left by blood culture analyser

Resources labs nº stickers, rack, scanner, LIMS, blood culture analyser

Description of task performed

Label request form labs nº
Label sample with labs nº
Put samples into rack
Scan request form.
Sample booked into Telepath LIMS.
Sample had electronic order.
Took samples over to analyser.
END of observation with G.
Q: In general samples are labelled ASAP and when labelled brought across to blood culture analyser.
Batches occur if more than sample is received.

Document: Observation form of Blood culture process Version: 1 Issue date: 08/Nov/2022

Figure S1 – Example observation form for blood culture processes. LIMS=laboratory information management system, MLA=medical laboratory assistant (technician)

Simulation input parameters (Table S1)

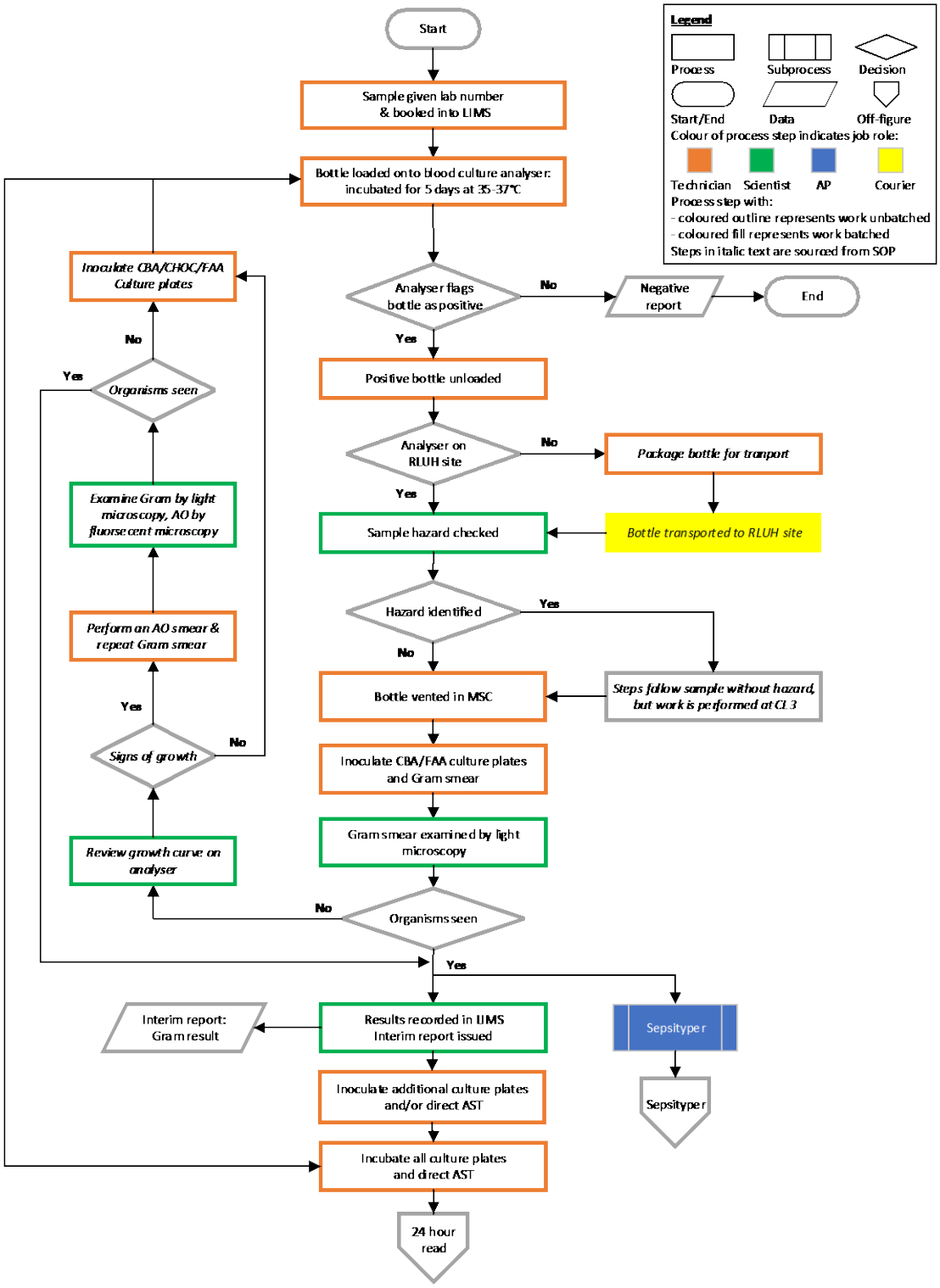
Table S1: Simulation input parameters not listed in main text.

Parameter name	Distribution/Value	Parameter values
Specimen generation rate	Exponential	0.0638
Shift working on	Boolean	True
Shift patterns (reception technician only)		08:00–20:00: On duty 20:00–08:00: Off duty
Time to positivity	Weibull	alpha = 1417; beta = 1.383
Maximum incubation time	Constant	5 days
Positivity rate	Proportion	11.1%
Transport time	Gamma	alpha = 0.99; beta = 333.3
AST incubation time*	Constant	18 hours
n specimens per simulation	Constant	500

*AST=antimicrobial susceptibility testing

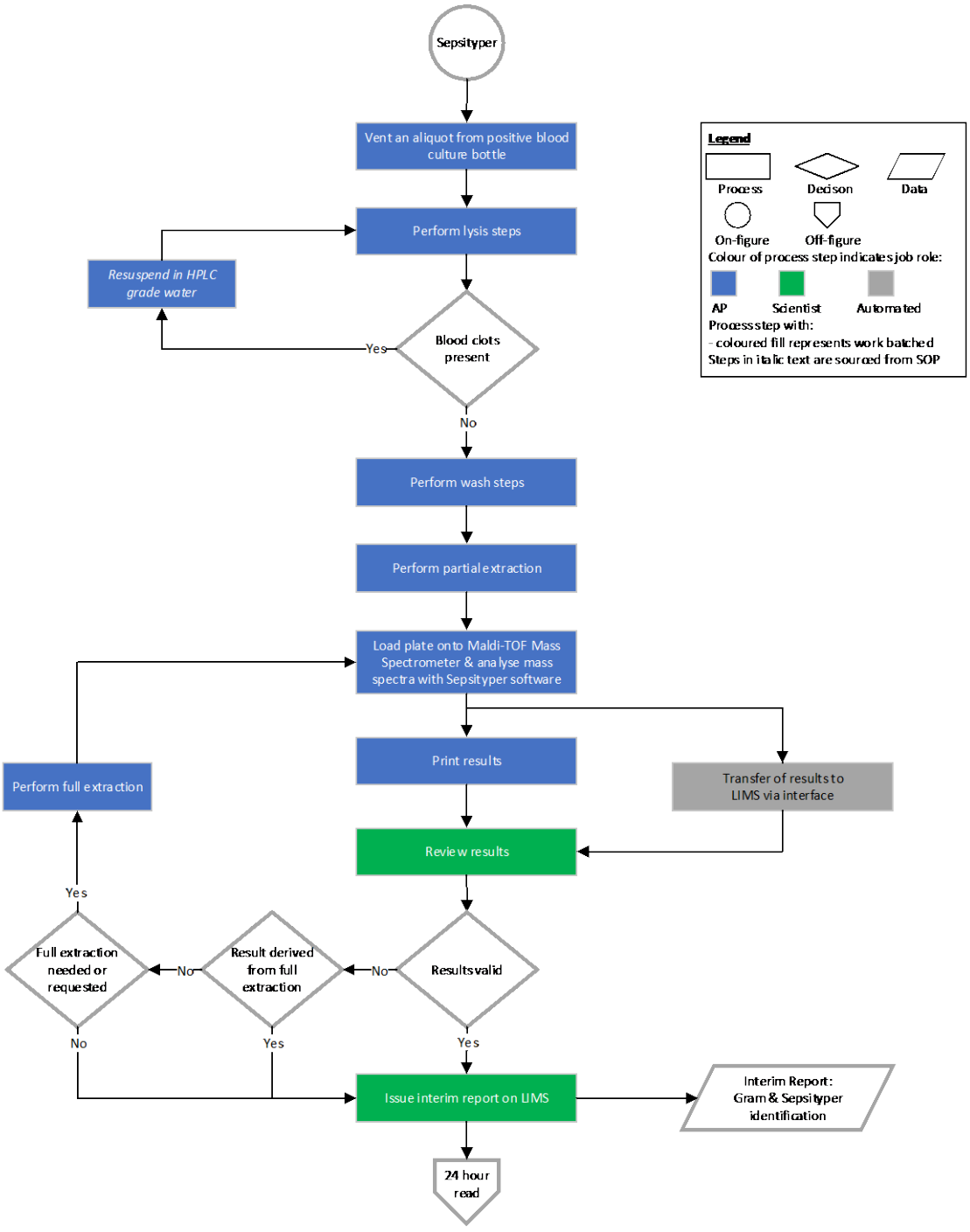
Blood culture process maps (Figure S2–Figure S5)

Note: Associate practitioners (AP) were classed as technicians within the simulator.



AP: Associate Practitioner; SOP: Standard Operating Procedure; LIMS: Laboratory Information Management System; RLUH: Royal Liverpool University Hospital; CBA: Columbia Blood Agar; CHOC: Chocolate Agar; FAA: Fastidious Anaerobe Agar; MSC: Microbiological Safety Cabinet; CL3: Containment Level 3; AO: Acridine Orange

Figure S2 – Process model of blood cultures from laboratory receipt to processing positive bottle.



AP: Associate Practitioner; SOP: Standard Operating Procedure; HPLC: High Performance Liquid Chromatography; MALDI-TOF: Matrix-Assisted Laser Desorption/Ionization Time-Of-Flight; LIMS: Laboratory Information Management System

Figure S3 – Process model of blood cultures for direct identification from positive bottle using Sepsityper.

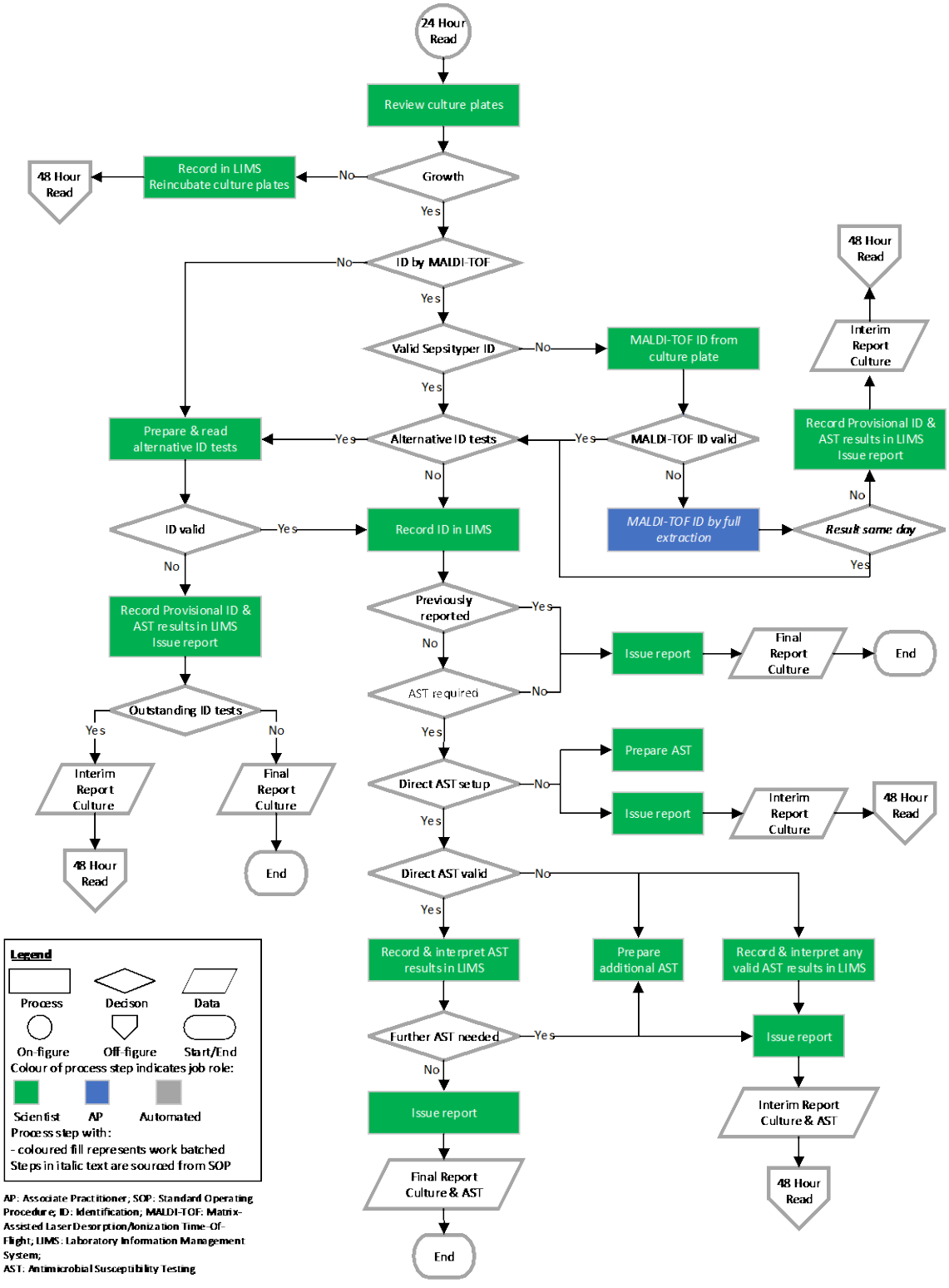


Figure S4– Process model of blood cultures for 24-hour culture read.

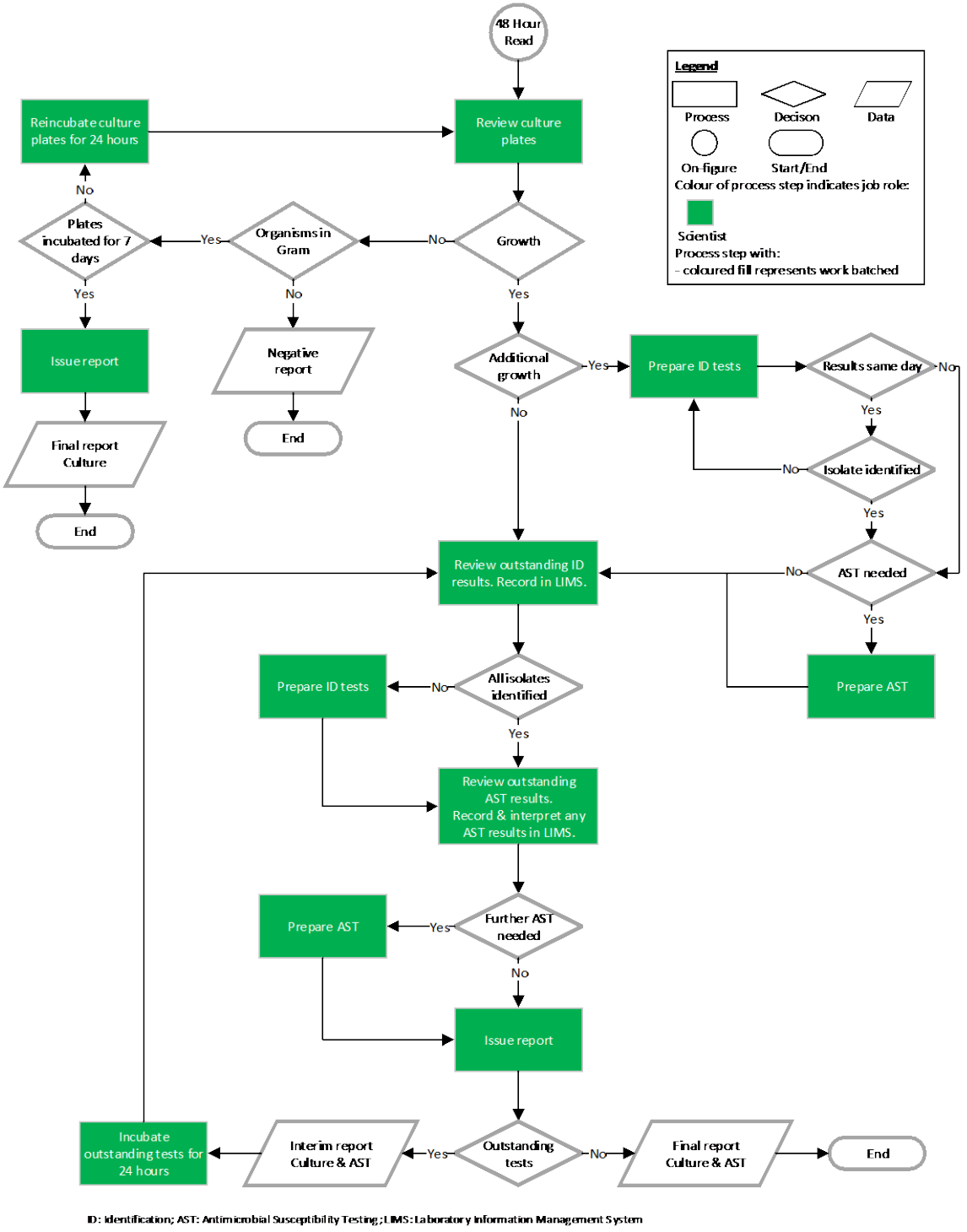


Figure S5 – Process model of blood cultures for 48 hour and post 48-hour culture reads.

Supplementary plots (Figure S6–Figure S8)

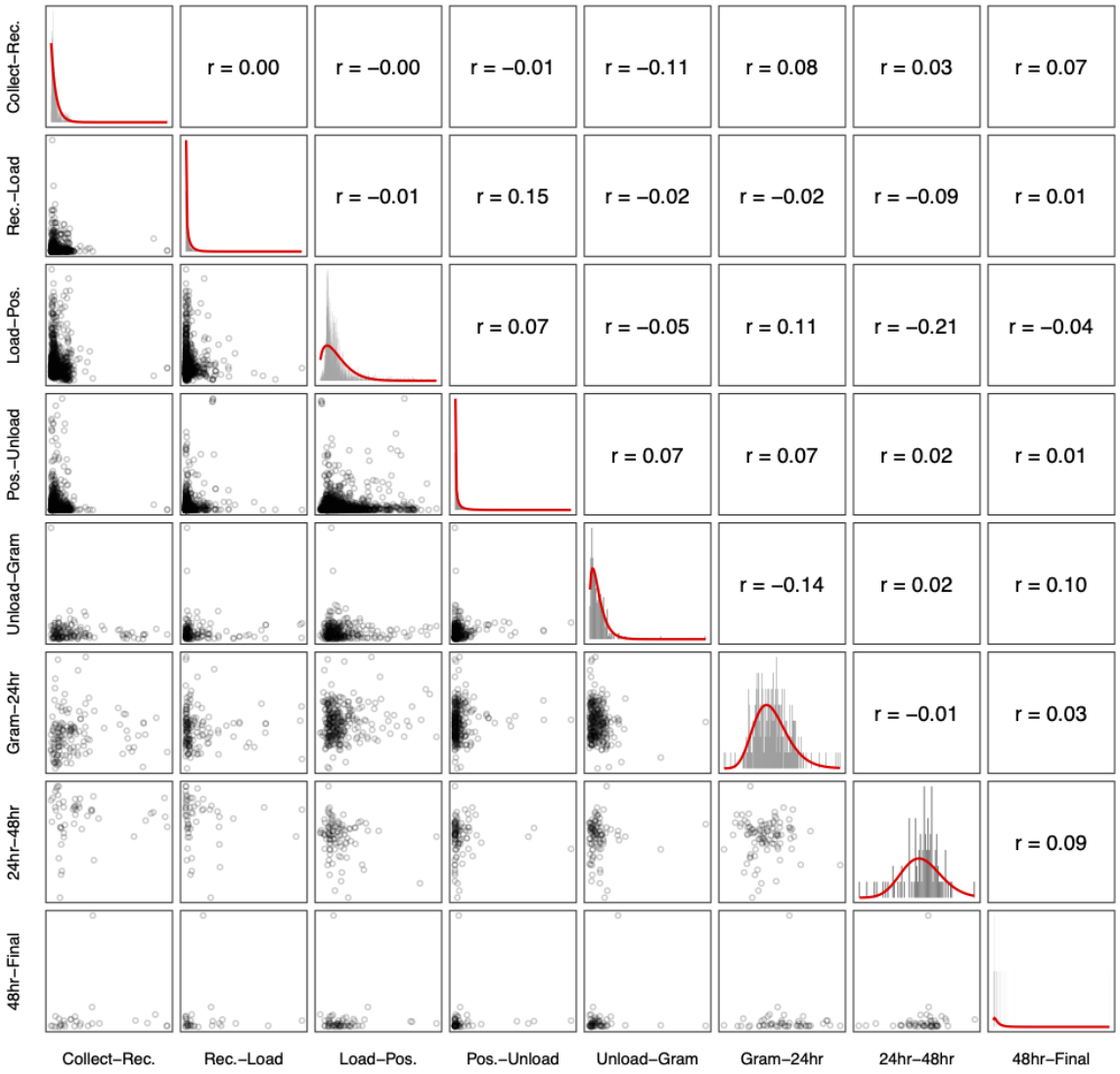


Figure S6 – Correlation matrix for retrospectively collected data. Diagonal blocks show data histograms and the chosen probability distributions. Lower blocks show data points. Upper blocks show Pearson correlation coefficient (r).

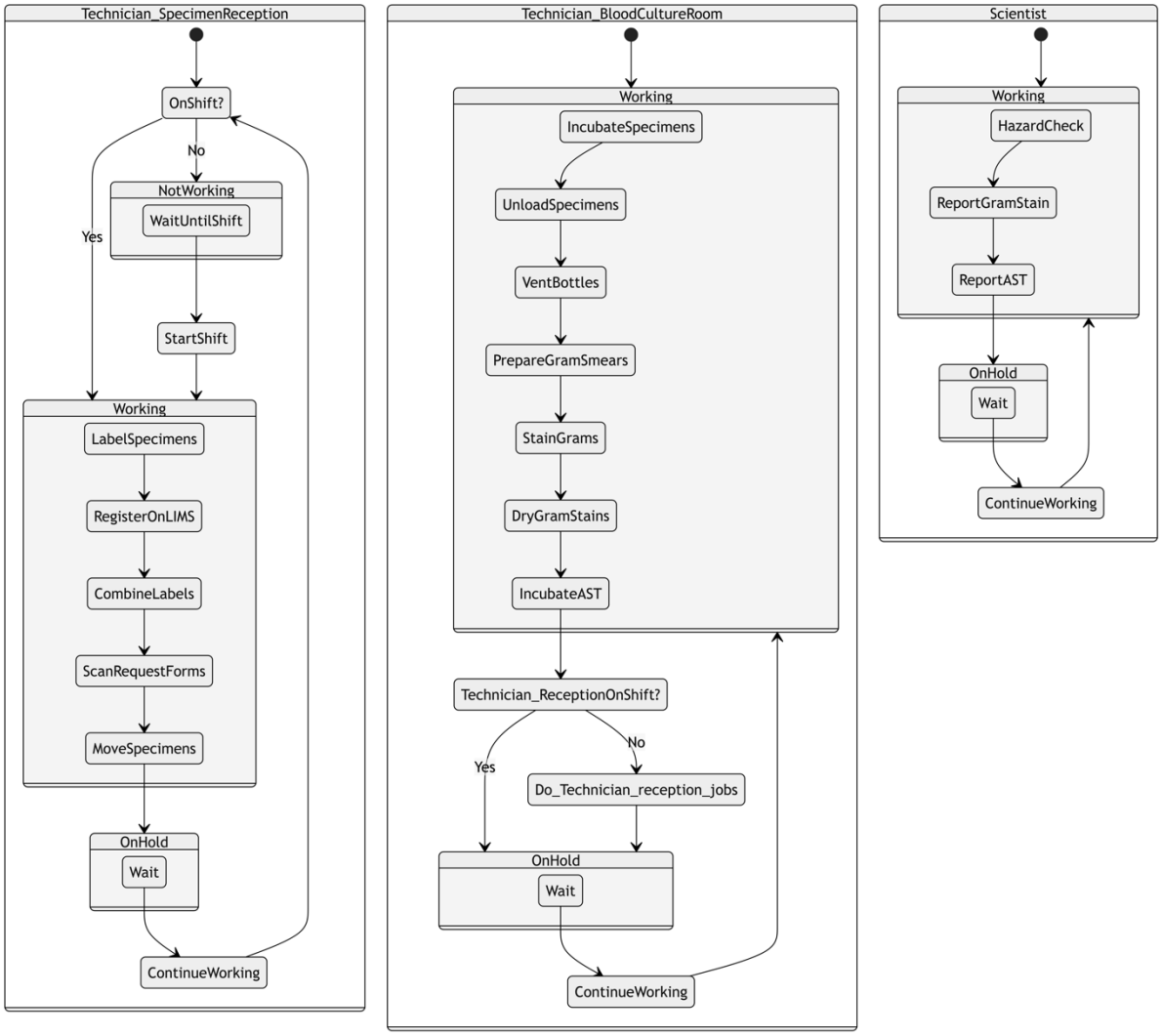


Figure S7 – Behavior of staff within the simulated laboratory. Technicians generally perform processing steps, while scientists generally perform interpretative steps. LIMS = laboratory information management system, AST = antimicrobial susceptibility testing.

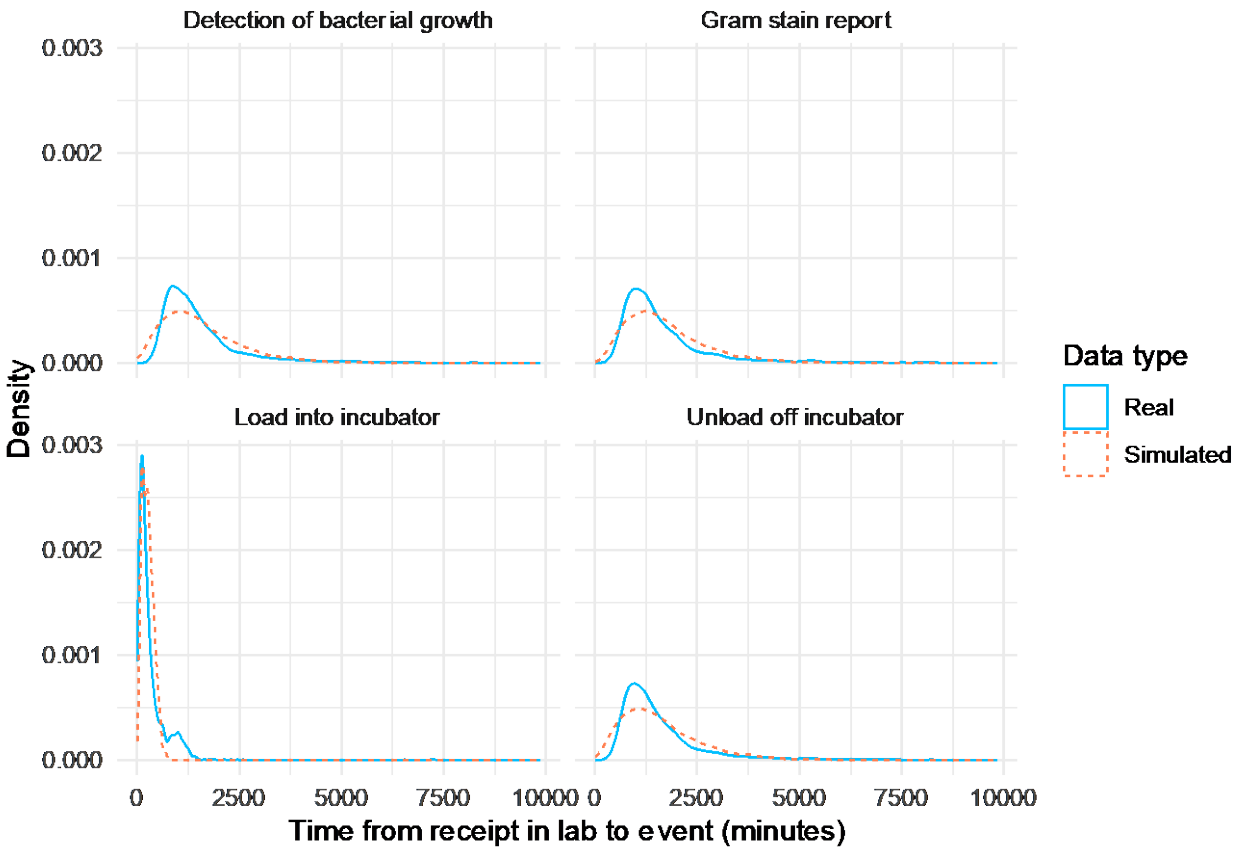


Figure S8 – Results of specimens generated from a single simulation run, with distribution of key event node times plotted against those observed in real specimens. 'OnHold' wait time was fixed at 23 minutes for all staff members.